

Deployment

Stages of
ML Pipeline

① Problem understanding



② Data collection



③ Data preprocessing



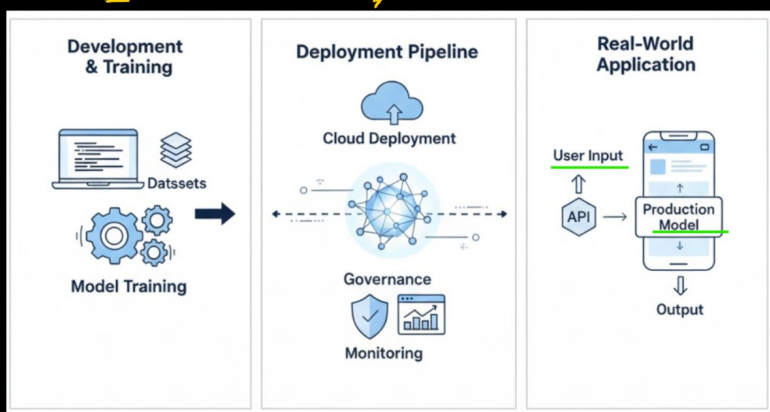
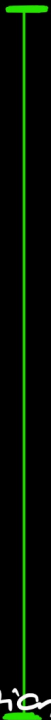
④ Model selection & Training



⑤ Model Accuracy/Evaluation



⑥ Model Deployment



Deployment is when the model leaves the "lab" and becomes part of a real product or service. At this point, the model must:

1. handle live data
2. work consistently
3. integrate with apps, APIs, or cloud services
4. respond quickly to user requests

This stage determines whether the model is practically useful, not just academically correct.

Governance is the long-term responsibility of monitoring and guiding the deployed model. It ensures the model behaves safely, ethically, and fairly over time.

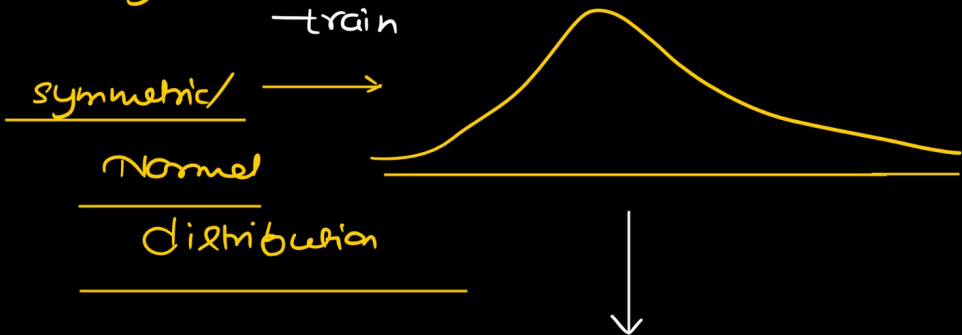
Governance focuses on:

- fairness
 - transparency
 - avoiding bias
 - preventing harmful behavior
 - ensuring the model still performs well
 - following legal and ethical guidelines ← medical data
- Unlike development, governance continues throughout the entire life of the model.

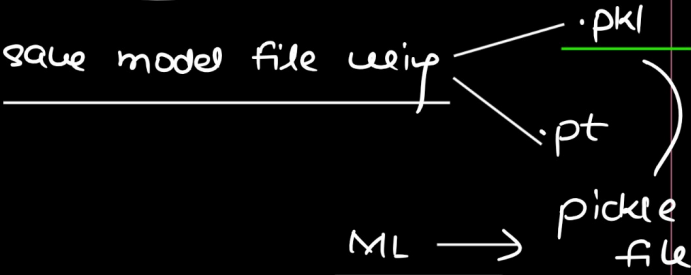
Stage	Purpose	Environment
<u>Model Development</u>	<u>Build the model</u>	<u>Offline, controlled</u>
<u>Deployment</u>	<u>Use the model</u>	<u>Real-world environment</u>
<u>Governance</u>	<u>Monitor and guide the model</u>	<u>Ongoing, long-term</u>

- * new user behavior it has never seen
- * shifting market trends
- * unexpected inputs →
- * changes in language or culture
- * noise or errors in real data

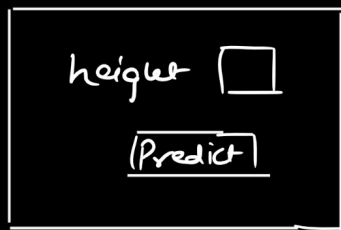
→ change in data distribution



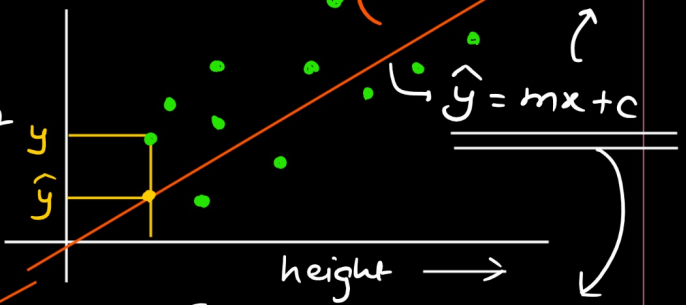
Weight, parameter, rule ← Model training



Linear Regression



fitness Tracker
detect movement
face unlock
weight



Deployment Strategy

Mobile phone

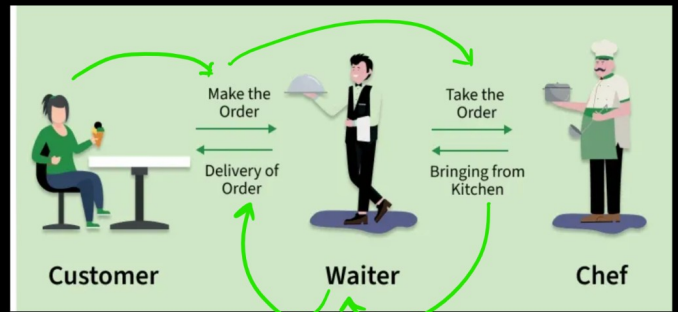
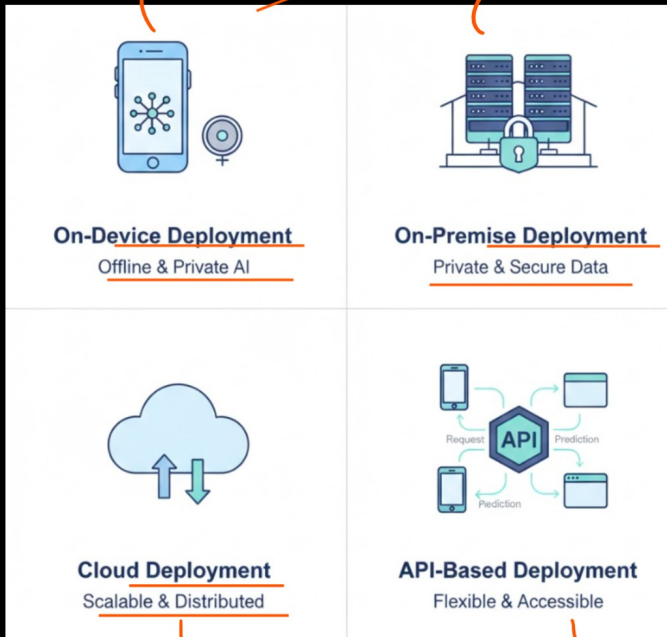
company's services

height →
to get the

Optimized

Value of
 m & c

hospitals,
bank
physical
servers
owned
by org

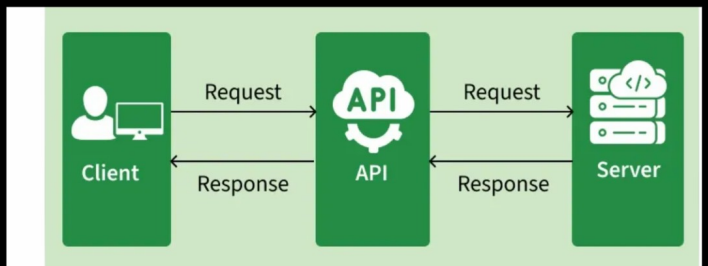


AWS, GCP,
Azure

(chatbots, Recommendation system,

GenAI
-tools)

(scalability,
Reliability,
flexibility)



cloud

Numeric

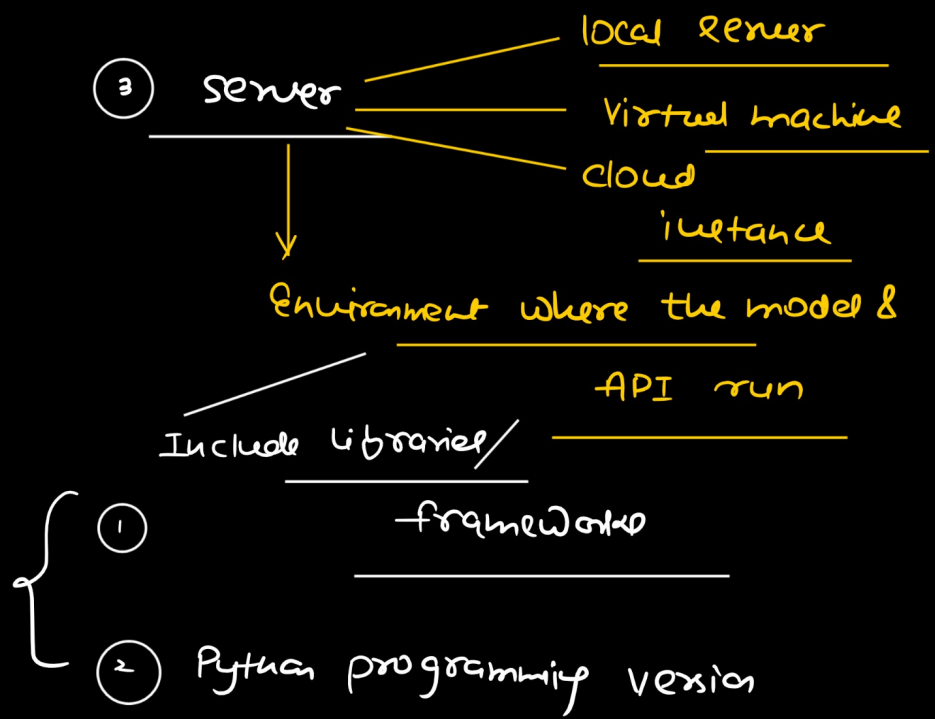
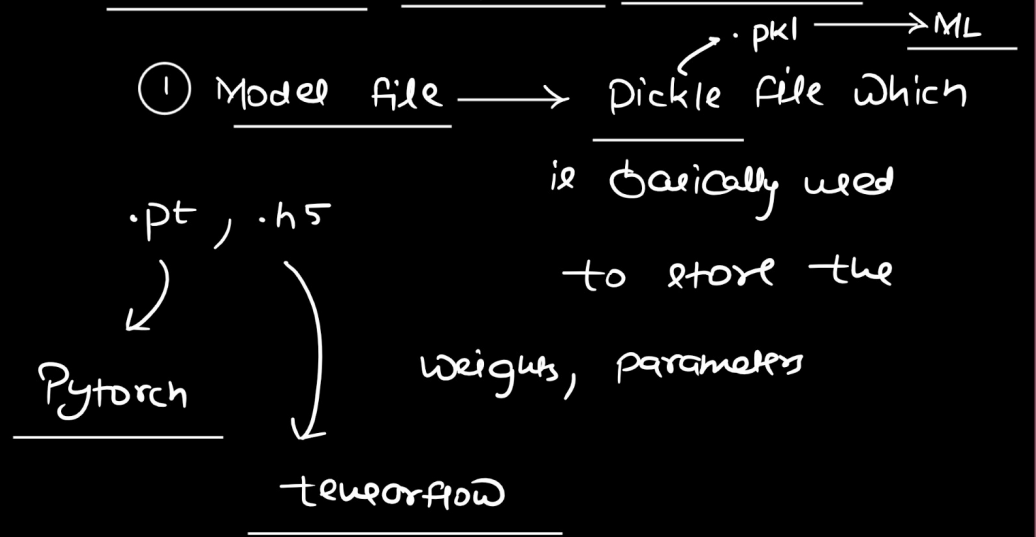
Data → ML → sklearn

Image
data

→ DL →

Pytorch
Research
or
Production
tensorflow

Core components of Deployed Model



Docker → model & all the
required file

Automation Tools

→ Jenkins → used to setup
pipeline that
run steps like
test the model,

GitHub Actions

Package the
model,
Automatically deploy the
run tests model
When code changes

Deploy model to
cloud services

Responsible AI

Responsible AI is the practice of developing and operating AI systems with the clear intention of preventing harm, reducing bias, protecting users, and ensuring lasting public trust.

Pillars of Responsible AI

